

Immunogenicity and protective efficacy of a tuberculosis DNA vaccine

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Tuberculosis is the most widespread and lethal infectious disease affecting humans. Immunization of mice with plasmid DNA constructs encoding one of the secreted components of *Mycobacterium tuberculosis*, antigen 85 (Ag85), induced substantial humoral and cell-mediated immune responses and conferred significant protection against challenge with live *M. tuberculosis* and *M. bovis* bacille Calmette-Guérin (BCG). These results indicate that immunization with DNA encoding a mycobacterial antigen provides an efficient and simple method for generating protective immunity and that this technique may be useful for defining the protective antigens of *M. tuberculosis*, leading to the development of a more effective vaccine.

The only tuberculosis vaccines currently available for human use are attenuated strains of *M. bovis*, termed bacille Calmette-Guérin (BCG), introduced more than 70 years ago. BCG continues to be widely administered to children in many countries, yet its efficacy remains controversial. As alternatives to BCG, there are several experimental approaches to tuberculosis prophylaxis, including rationally attenuated *M. tuberculosis*, recombinant BCG overexpressing *M. tuberculosis* antigens, protein subunit vaccines and DNA (nucleic acid) vaccines.

Administration of plasmid DNA expression vectors has been shown to result in protein expression *in vivo*, the generation of humoral and cell-mediated immune responses, and protective efficacy in animal models of infectious disease¹. Specifically, administration of DNA encoding influenza hemagglutinin and nucleoprotein conferred protective humoral and cross-strain cellular immunity, respectively, in mice². Subsequently, DNA vaccines were shown to be effective for inducing immune responses and/or protection in other animal models, including HIV in mice, influenza in chickens and ferrets, bovine herpesvirus in mice and cattle, rabies virus in mice, malaria in mice, leishmaniasis in mice, herpes simplex virus in mice, and cottontail papilloma virus in rabbits (for review see ref. 1). DNA vaccines also have been used to generate immune responses against a variety of viral, parasite and tumor antigens in animals ranging from mice to nonhuman primates. From these preclinical studies, DNA vaccination seems to be a broadly applicable technique for generating protective immune responses against infectious pathogens. DNA vaccines induce such responses without the need for live organisms, replicating vectors or adjuvants and, therefore, may offer significant practical and safety advantages over certain other

approaches. Potential safety considerations pertaining to DNA vaccines, such as insertional mutagenesis, autoimmune responses and immunologic tolerance, are currently being investigated in detail.

The protective antigens for tuberculosis are still not precisely defined, but the prevailing hypothesis that they predominantly reside within the secreted or exported proteins of the bacillus has been recently supported by observations in animal models³. The three components of the Ag85 complex, a 30- to 32-kilodalton family of proteins (Ag85A, Ag85B and Ag85C), constitute a major fraction of the secreted proteins in the culture filtrate (CF)⁴. Ag85 homologues are found in all mycobacterial species and in *Corynebacterium* sp., but not in other related intracellular pathogens such as *Nocardia* or *Listeria*. The Ag85 complex induces strong T-cell proliferation, interferon- γ (IFN- γ) production and cytotoxic T lymphocyte (CTL) activity in most healthy individuals infected with *M. tuberculosis*/*M. leprae*⁵. Sensitization against Ag85 occurs in the first phase of mycobacterial infection⁶. Hence, recognition of these antigens may be an effective strategy of the host to control initial bacterial replication, consequently making the proteins of the Ag85 complex reasonable vaccine candidates.

In the present study, we show that the DNA vaccine approach is well suited to the development of an effective tuberculosis vaccine because of its ability to generate robust T-cell responses against the antigen 85 complex and protection in mouse challenge models. Furthermore, the phenotypes of the humoral and cellular immune responses induced are indicative of a Th1-type of T-cell response, which is generally believed to be necessary for protection against mycobacterial disease.

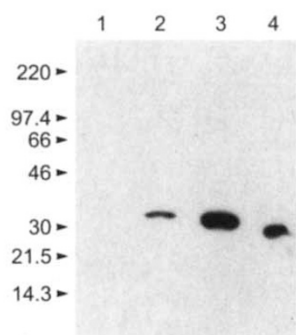


Fig. 1 Expression and secretion of Ag85 in eukaryotic cells. RD cells were transfected with plasmids encoding secreted Ag85A (lanes 2, 3) and immunoblot analysis was performed with Moab 17-4 (diluted 1:40). Culture supernatants (lane 2) and cell lysates (lane 3) were analyzed in this way. Molecular mass markers are noted on the left side of the figure. Lane 1 contains RD cells transfected with control DNA as a negative control, and lane 4 contains BCG CF as a positive control.

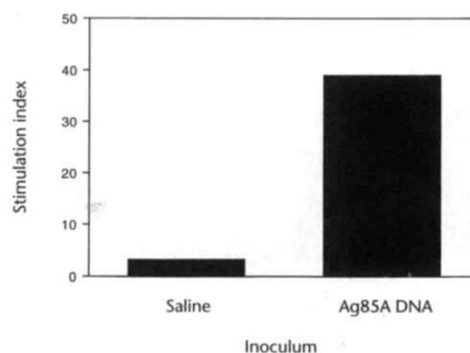


Fig. 2 Induction of lymphoproliferative responses by Ag85 DNA. Pools of spleen cells ($n = 3$) taken from BALB/c mice 28 days after the last vaccination with secreted Ag85A DNA (three doses of 100 μ g) were stimulated *in vitro* with BCG CF. Data are expressed as mean stimulation index for pools of spleen cells from 3 mice tested in triplicate. As a negative control, spleens from mice injected with saline were similarly processed.

In vitro expression of Ag85

Plasmid DNA vectors were constructed to encode a secreted form of Ag85A, by using the signal sequence from tissue plasminogen activator, and a mature form of Ag85A not containing a signal sequence. In order to ensure that the plasmid DNA vaccine constructs were intact and functional, the plasmids were sequenced across the gene inserts and expression was analyzed *in vitro* by transient transfection. A protein of ~35 kDa was expressed in and secreted by rhabdomyosarcoma (RD) cells transfected with secreted Ag85A DNA, as detected by immunoblot using a monoclonal antibody, but no immunoreactive proteins were detected in cells transfected with control DNA (plasmid not containing a gene insert) (Fig. 1). The secreted Ag85 expressed in RD cells migrated on the gel at a slightly higher molecular mass compared with native Ag85 proteins present in BCG CF, because of the presence of an N-linked oligosaccharide in the eukaryotically expressed protein (A.M.Y. *et al.*, unpublished observations). Cells transfected with mature Ag85A expressed an ~33-kDa protein similar in electrophoretic mobility to native Ag85 (data not shown).

Humoral immune responses

Using an enzyme-linked immunosorbent assay (ELISA), high-titer antibodies reactive against Ag85 were measured in sera from mice immunized with Ag85A DNA (Table 1). No Ag85-specific antibodies were detected in mice vaccinated with control DNA. Mice vaccinated with mature Ag85A DNA (encoding Ag85A without a signal sequence) produced slightly lower antibody titers (not shown). Antibody isotypes were predominantly IgG1 and IgG2a, but the IgG2b isotype was also detected (particularly in B6 mice). The IgM and IgG3 isotypes could not be detected.

Cell-mediated immune responses

In order to investigate cell-mediated immune responses induced by Ag85 DNA vaccines, spleen cells from vaccinated mice were restimulated with antigen *in vitro*. Substantial lymphocyte proliferation was observed in response to BCG CF in spleen cells from BALB/c mice vaccinated with Ag85A DNA but not with saline (Fig. 2) or with control DNA (data not shown). Significant proliferation was also seen when spleen cells were restimulated

in vitro with purified, native Ag85 and in spleen cells from mice vaccinated with mature Ag85A DNA (data not shown).

Spleen cells from BALB/c and B6 mice vaccinated with Ag85A DNA produced elevated levels of interleukin-2 (IL-2) and IFN- γ upon restimulation with purified protein derivative (PPD) or purified Ag85 (Fig. 3). The levels of cytokine secretion approached those from cells nonspecifically stimulated with mitogen (data not shown). In addition, splenocytes from DNA-vaccinated mice reacted against the same immunodominant Ag85 peptides as during a natural BCG infection. Specifically, splenocytes from B6 mice were particularly reactive against 20-amino acid peptides previously shown to be H-2^b-restricted T-cell epitopes corresponding to amino acids 241–260 (Fig. 3, *a* and *c*) and amino acids 261–280 of Ag85A (data not shown). *In vitro* responses to whole BCG bacilli and BCG cytoplasmic extract were only minimally induced in spleen cells from mice vaccinated with Ag85A DNA (not shown). Production of tumor necrosis factor- α (TNF- α) and granulocyte-macrophage colony-stimulating factor (GM-CSF) were also significantly increased, whereas little to no detectable levels of IL-4, IL-6 and IL-10 were found in supernatants from antigen-stimulated spleen cell cultures from Ag85A DNA-vaccinated mice (Table 2).

Ag85-specific CTL activity was demonstrated in spleen cells from Ag85A DNA-vaccinated BALB/c mice that were nonspecifically activated with concanavalin A (conA) and IL-2. Significant lysis of Ag85A-expressing transfected P815 mastocytoma target cells, but not mock-transfected cells, was observed (Fig. 4*a*). Ag85-

Table 1 Humoral immune responses to Ag85 by DNA vaccination

	DNA	Total	IgM	IgG1	IgG2a	IgG2b	IgG3
BALB/c	Control	0.48	20	40	20	20	20
	85A	10.6	20	5120	2560	640	20
C57BL/6	Control	0.15	20	40	20	20	20
	85A	47.6	20	1280	2560	5120	20

Serum samples from 15 mice per group were pooled and tested in an ELISA. Total anti-Ag85 immunoglobulin is expressed as micrograms/milliliter and levels of specific immunoglobulin classes as reciprocal end-point titer.

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specific CTLs were also demonstrated in antigen-restimulated spleen cells from Ag85 DNA-vaccinated mice. In this case, lytic activity was determined to be mediated by CD8⁺ T cells, as indicated by inhibition of lysis after treatment of spleen cells with anti-CD8 antibodies and complement (Fig. 4b). Treatment of cells with anti-CD4 antibodies and complement did not result in a significant reduction in specific lysis (20% lysis versus 25% in control cells at an effector:target ratio of 25:1). In a separate set of experiments, CTL activity against Ag85-pulsed peritoneal exudate cells was demonstrated by splenocytes from B6 mice vaccinated with Ag85A DNA (not shown).

Protective efficacy in animal models

To measure Ag85 DNA vaccine efficacy, mice were challenged by intravenous (i.v.) administration of live *M. bovis* BCG or by aerosol administration of *M. tuberculosis*, and mycobacteria were quantified in the spleens and lungs after infection. As shown in Fig. 5a, vaccination with secreted Ag85A DNA conferred protection from *M. bovis* BCG replication compared with vaccination with control DNA ($P < 0.005$ on day 9 after challenge), reducing replication by a factor of 20 to 40, similar to that induced by the BCG vaccine. Vaccination with DNA constructs encoding the mature form of Ag85A likewise reduced the number of colony-forming units (CFUs) in the lungs of challenged mice ($P < 0.025$ in B6 mice; $P < 0.05$ in BALB/c mice) (data not shown).

Ag85A DNA also conferred protective efficacy in a low-dose aerosol *M. tuberculosis* challenge. At one month after challenge, significant reductions in CFU in the lungs of mice vaccinated with

Table 2 Cytokine secretion induced *in vitro* by spleen cells of Ag85 DNA-vaccinated mice

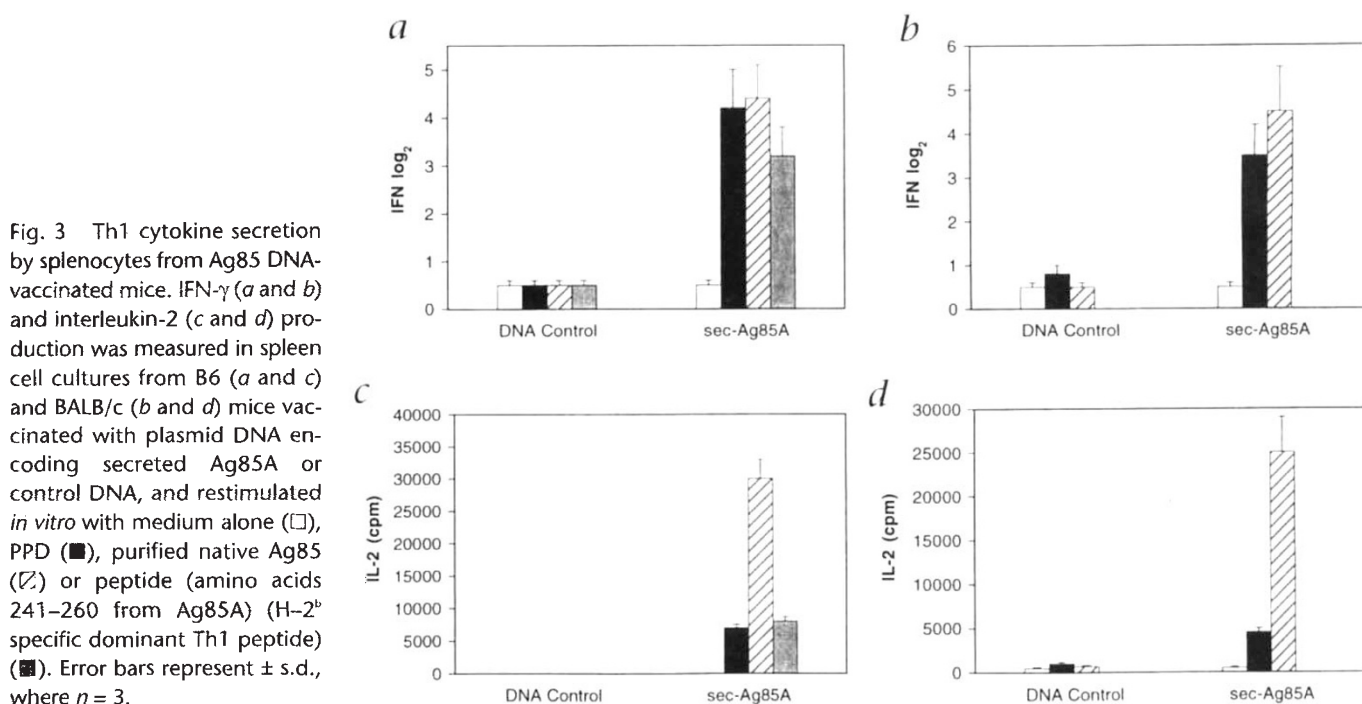
	DNA	IL-4		IL-6		IL-10		TNF- α		GM-CSF	
		Ag85	PWM	Ag85	BCG	Ag85	PWM	Ag85	BCG	CF	ConA
BALB/c	Control	<15	514	47	184	<100	1755	23	1951	7.4	213
	85A	<15	664	101	429	<100	297	162	2031	119	388
C57BL/6	Control	<15	122	19	171	<100	351	6	2084	ND	ND
	85A	<15	184	10	50	<100	1109	370	2145	ND	ND

Spleen cells from three mice were pooled and restimulated *in vitro* with the indicated antigen or mitogen. Samples were tested in triplicate and levels presented as mean picogram/milliliter with <10% variance. (ND, not determined; PWM, pokeweed mitogen.)

Ag85A DNA were observed compared with those of mice injected with saline or control DNA ($P < 0.05$) (Fig. 5, b and c). This was true for DNA constructs encoding secreted Ag85A, mature Ag85A or a combination of the two. The data shown indicate that this protection induced by Ag85A DNA persisted for at least 70 days after vaccination ($P < 0.005$) (Fig. 5c). The levels of protection induced by Ag85A DNA vaccines (8- to 16-fold) were comparable to that induced by a BCG vaccine under similar conditions (Fig. 5d).

Discussion

Protective immunity against mycobacterial infection is mediated by interactions between specifically sensitized CD4⁺ and CD8⁺ T lymphocytes and activated macrophage effector cells harboring the intracellular pathogen⁷. These cells interact with one another by means of a complex network of cytokines. Th1 helper T cells are thought to be essential, because their cytokines (for example, IL-2 and IFN- γ) are predominantly secreted in individuals resistant to mycobacterial infection (that is, healthy humans with *M. tuberculosis* and *M. leprae* primary infection and tuberculoid leprosy patients)⁸. In contrast, Th2 helper T-cell cytokines (for example, IL-4 and IL-10) have been found mainly in susceptible individuals (that is, tuberculosis patients with extensive lung lesions, lepromatous



leprosy patients and BCG-infected BALB/c mice)^{9,10}. Optimal proliferation of Th1 and Th2 subsets is thought to depend on distinct antigen-presenting cells and their costimulatory factors, such as IL-12 and IL-10, respectively. Furthermore, $\gamma\delta$ T cells also may influence the initial activation of these two T-cell subsets. IFN- γ , a potent activator of macrophages¹¹ and produced by both CD4⁺ and CD8⁺ T cells, is one of the crucial cytokines in antimycobacterial protection, as demonstrated in knock-out mice that have had either the IFN- γ or the IFN- γ receptor genetically deleted^{12,14}. TNF- α is also required for protection against *M. tuberculosis* in mice¹⁵, and GM-CSF has been shown to have mycobacteriostatic properties when added to macrophages infected with *M. avium*¹⁶ or *M. tuberculosis*¹⁷. Finally, CD8⁺ CTLs also play an important role in protection against mycobacterial disease¹⁸.

Here we show that DNA vaccination with plasmids encoding mycobacterial genes is a potent method for the generation of specific helper responses with a Th1-like phenotype, and CTL responses against protein antigens from *M. tuberculosis*. In the past few years, DNA vaccination has been reported by many authors to be an effective strategy for expression of foreign antigens *in vivo*, leading to immunization against several different viruses and protozoa¹ and suggesting that this approach may be generally applicable and could potentially be applied to mycobacterial disease. In support of this possibility was the observation that vaccination of mice with cells expressing heat-shock protein

hsp65 could be protective in an *M. tuberculosis* challenge model¹⁹. The expression of Ag85 in RD cells *in vitro* after transfection and the generation of antibody responses *in vivo* shown here indicate that mycobacterial proteins can be efficiently expressed in eukaryotic cells *in situ* after DNA injection. Furthermore, induction of helper T cells and CTLs demonstrates that these proteins can be processed and presented by MHC class I and II molecules *in vivo*.

Protective antigens for tuberculosis (and leprosy) are ill-defined at the present time, in part because effective vaccination methods to delineate such antigens and specific adjuvants for the generation of strong cellular immunity are not readily available, and also because classical purification of mycobacterial antigens is time-consuming and difficult. The technique of DNA vaccination

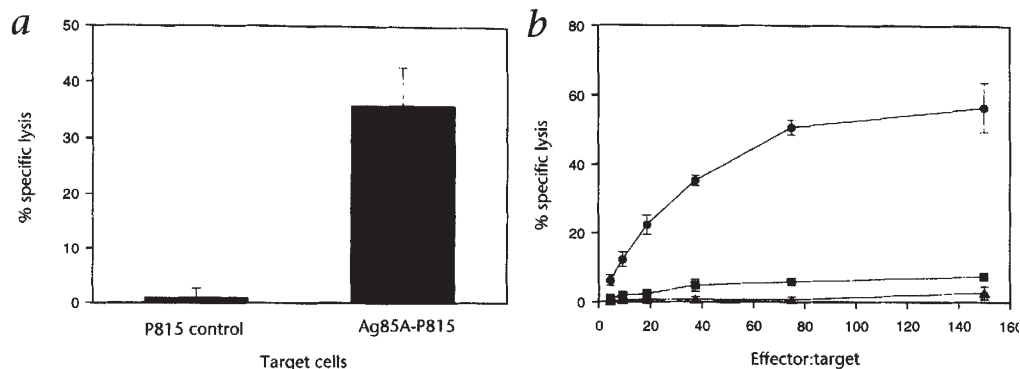
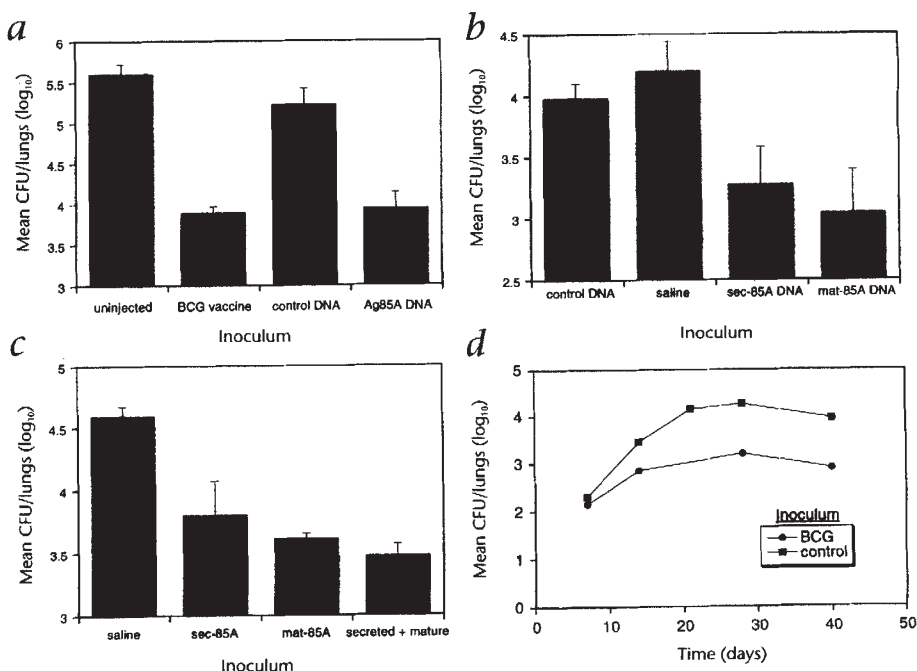


Fig. 4 Cytotoxic T lymphocyte responses by splenocytes from Ag85 DNA-vaccinated BALB/c mice. **a**, CTL activity in spleens from BALB/c mice vaccinated with secreted Ag85A DNA ($3 \times 100 \mu\text{g}$). Splenocytes were nonspecifically activated *in vitro* with IL-2 and conA in the absence of Ag85, then CTLs were assessed using P815 target cells stably transfected with secreted Ag85A DNA (Ag85A-P815). Untransfected P815 were used as controls. Effector:target ratio was 100:1, and error bars represent $\pm$ s.d., where $n = 3$. **b**, CTL activity of spleen cells from BALB/c mice vaccinated with DNA encoding mature Ag85A ($3 \times 100 \mu\text{g}$). Spleen cells were restimulated *in vitro* with Ag85-transfected P815 cells and lytic activity was measured against untransfected P815 cells (squares) or Ag85-transfected P815 cells using undepleted (circles) and CD8-depleted spleen cells (triangles) at the indicated effector:target ratios. Data are presented as percent specific lysis $\pm$ s.d., where $n = 3$.

Fig. 5 Protective efficacy of Ag85 DNA vaccines. **a**, Protection of C57BL/6 mice from BCG challenge. Mice were vaccinated with control DNA, secreted Ag85A DNA ($3 \times 100 \mu\text{g}$), or BCG vaccine or were uninjected then challenged with BCG and the number of CFU/lungs was measured 9 days later. Data are presented as $\log_{10}$ CFU $\pm$ s.d., where $n = 3-4$. **b-d**, Protection of C57BL/6 mice from aerosol *M. tuberculosis* challenge. Mice were vaccinated with secreted Ag85A DNA (sec-85A) or mature Ag85A DNA (mat-85A), or a combination of both, then challenged with *M. tuberculosis* 28 days (**b**) and 70 days (**c**) after vaccination. The number of CFU/lungs was measured one month later. As negative controls, mice were untreated or vaccinated with control DNA or saline. As a positive control, age-matched mice received the BCG vaccine and CFU in the lungs were measured at various times after aerosol challenge (**d**). Data are presented as $\log_{10}$ CFU $\pm$ s.d., where $n = 4$.



could offer a solution to some of these problems, by providing an effective means of inducing relevant cytokines known to be involved in immunity (that is, IFN- γ , IL-2, GM-CSF, TNF- α) and CTL responses, without the need for adjuvants or prior protein purification. Mice vaccinated with Ag85A DNA had a higher capacity to control mycobacterial replication than mice injected with saline or control DNA, which is consistent with the hypothesis that the Ag85 complex is involved in immune protection and which corroborates recent work demonstrating that a vaccine consisting of purified Ag85 protein and an adjuvant protected to some extent guinea pigs from *M. tuberculosis* challenge²⁰. Interestingly, Ag85A DNA-vaccinated mice challenged with BCG showed a rapid decrease in CD45RB expression on their CD4⁺ T cells (K.H. *et al.*, unpublished observations), comparable to the previously reported downregulation observed in memory-immune, *M. tuberculosis*-challenged mice²¹. This implies that memory effector cells were generated by the DNA vaccination, although this still remains to be determined in animal models.

In conclusion, this study demonstrates that DNA vaccination using plasmids encoding a single mycobacterial gene is an effective means of generating strong Th1 and other relevant cytokines, such as GM-CSF and TNF- α , and CTL responses leading to protective efficacy in animal models of disease. The level of this protection could potentially be enhanced by the inclusion of DNA plasmids encoding other mycobacterial antigens, since other secreted proteins are known to induce strong T-cell responses. Furthermore, DNA vaccination may provide a simple means to screen as-yet-unidentified mycobacterial antigens for the development of a more effective tuberculosis vaccine, through the use of approaches such as genome sequencing and expression library immunization²².

Methods

Plasmid construction. The Ag85A gene from *M. tuberculosis* was amplified from plasmid p85A.tub²³ in BlueScribe M13⁺ (VCS/Stratagene) by PCR with *Bgl*II site containing primers. Briefly, PCR was performed in a 100- μ l volume, containing 2.5 U cloned *Pfu* DNA polymerase (Stratagene), 200 mM dNTP, 0.5 μ g of each primer and 250 ng template DNA in the reaction buffer supplied with the enzyme (Stratagene). A Hybaid thermal reactor was programmed as follows: 5 min denaturation at 94 °C followed by 25 cycles (1 min at 94 °C, 2 min at 55 °C and 3 min at 72 °C) ending with 10-min extension at 72 °C. Amplified DNA was digested with Proteinase K (Boehringer Mannheim), heated 10 min at 95 °C followed by two phenol:chloroform:isoamyl alcohol extractions and precipitated with 1 volume of isopropanol. DNA was digested with *Bgl*II, isolated on a 1% agarose gel and extracted on "Prep a gene" (Bio-Rad). Fragments were cloned into the *Bgl*II site of the V1Jns.tPA vector²⁴, as previously described²⁵. As a control, the V1Jns vector not encoding a gene insert was used ("control DNA").

In vitro expression of Ag85. Rhabdomyosarcoma (RD) cells (ATCC CCL 136, American Type Culture Collection) were grown in high-glucose DMEM supplemented with 10% heat-inactivated fetal calf serum, 2 mM L-glutamine, 20 mM HEPES, 100 U/ml penicillin, and 100 μ g/ml streptomycin (Gibco BRL), and transfected with 10 μ g Ag85 DNA together with 10 μ g chloramphenicol acetyltransferase (CAT) DNA according to CellPfect kit instructions (Pharmacia). Cultures were glycerol-shocked for 2.5 min, 5 h after addition of calcium phosphate, refed and incubated for 2–3 days before harvest. Expression of CAT was quantified by an ELISA according to kit instructions (Boehringer Mannheim). Expression of Ag85 was visualized by immunoblot on Immobilon-P membranes (Millipore)

probed with monoclonal antibody 17-4 (ref. 26).

To generate stably transfected cells expressing Ag85, P815-HTR were transfected using Lipofectamine (Gibco) and the cesium chloride purified plasmids for secreted Ag85A DNA or mature Ag85A DNA, along with a plasmid for drug resistance to G418 (pcDNA3, Invitrogen). Clonal isolates of drug-selected cells were prepared by limiting dilution in 96-well plates and analyzed for Ag85 expression.

Vaccination. BALB/c and C57BL/6 (B6) mice (6- to 8-week-old females) purchased from Iffa-Credo, Bantin and Kingman, Charles River Laboratories and Jackson Laboratories, were anesthetized by intraperitoneal injection of ketamine/xylazine (100 mg/kg and 10 mg/kg, respectively), and injected i.m. three times (at 3-week intervals) in both quadriceps with 100 μ g Ag85A DNA using a 0.3-cc insulin syringe (Becton Dickinson). As negative controls, mice were uninjected, injected with saline or with control plasmid DNA (not containing a gene insert) in saline. As a positive control, mice were injected i.v. with 0.5 mg BCG.

Antigens. Live *M. bovis* BCG bacilli (BCG) were prepared from a 2-week-old surface pellicle culture of *M. bovis* BCG, strain GL2. Concentrated CF was obtained by ammonium sulfate precipitation of the culture supernatant of these BCG cultures²⁷. PPD was prepared at the Pasteur Institute from 8-week-old CF of strain *M. bovis* Vallée. Soluble cellular extract (extract) was prepared from *M. bovis* BCG GL2 by disruption of the cells in a French press, followed by centrifugation at 10,000g for 30 min and 100,000g for 90 min at +4 °C. The extracellular 30- to 32-kDa antigens (Ag85 complex) were purified from *M. bovis* BCG CF as described previously²⁸. CF, purified protein derivative (PPD), BCG bacilli and BCG extract were used at a final concentration of 25 μ g/ml, the purified Ag85 at 5 μ g/ml. Synthetic peptides were prepared as described before²⁶ and used at final concentrations of 10 μ g/ml. Polyclonal T-cell activation was measured with pokeweed mitogen (PWM, Gibco BRL, 1:50 final dilution) or concanavalin A (conA, Sigma, 4 μ g/ml).

ELISA. Levels of anti-Ag85 antibodies were determined by ELISA of pooled sera from fifteen mice as described previously²⁹. Serum titer was converted to antibody concentration by comparison with standard, and mean antibody concentration was calculated from three points of the linear portion of the titration curve. For isotype analysis, peroxidase-labeled, rat anti-mouse IgM, IgG1, IgG2a, IgG2b and IgG3 (Experimental Immunology Unit, Université Catholique de Louvain) were used.

Lymphoproliferation. Single-cell suspensions of spleen cells from DNA-vaccinated animals were depleted of erythrocytes in ACK lysis buffer (B&B Research Laboratories) and stimulated with BCG CF or purified, native Ag85 (2 μ g/ml) *in vitro* in round-bottomed microwell plates at 5×10^5 cells/ml in RPMI 1640 medium, supplemented with HEPES, glutamine, 10% FCS and 50 μ M 2-mercaptoethanol. Cells were cultured for 5 days, and [³H]thymidine was added at 1 μ Ci/well during the last 24 h. Cells were harvested onto glass fiber filter mats using a Titertek cell harvester and radioactivity was measured in a liquid scintillation counter (Betaplate, Wallac).

Cytokine production. Three weeks after the last DNA injection, three mice per group were killed, their spleens were removed and single-cell suspensions were prepared. Supernatants from three separate wells were pooled for each animal, stored frozen at -20 °C until assay, and tested at least twice. IL-2, IL-4, IL-10, GM-CSF, IFN- γ , and TNF- α levels were measured by an ELISA according to kit instructions (Endogen

and Genzyme); IL-2, IL-6 and IFN- γ levels were also measured using the following bioassays. To measure IL-2, 100 μ l of 24-h culture supernatant was added to 100 μ l of CTLL-2 cells (10^5 /ml) and incubated for 48 h, and [3 H]thymidine (8.3 Ci/ml, Amersham) was added (0.4 μ Ci/well) during the last 6 h of culture¹⁰. IL-2 levels are expressed as mean counts per minute $\pm$ s.d. obtained in three mice. Samples were compared to a reference standard²⁶ and the limit of detection was 50 pg/ml. For IFN- γ , antiviral activity in 72 h culture supernatants was measured in duplicate using a cytopathic effect reduction assay of Vesicular Stomatitis Virus on mouse L929 cells, as described elsewhere¹⁰. Titers were calculated as mean log₂ $\pm$ s.d. obtained in three mice. In this assay, one log₂ unit corresponds to 4 IU/ml or 1360 pg/ml and detection limit is >2 IU/ml or >680 pg/ml. IL-6 activity was assessed in triplicate in a colorimetric assay by measuring hexose-aminidase levels of 7TD-1 mouse-mouse hybridoma cell cultures grown in absence or presence of serial fivefold dilutions of 72-h culture supernatants¹⁰. IL-6 titers are expressed in mean laboratory units per milliliter. One unit corresponds to approximately 1 pg IL-6.

Cytotoxic T lymphocyte activity. Spleen cells from DNA-vaccinated BALB/c mice were restimulated *in vitro* with conA and IL-2, as described before², or with IL-2 (10 U/ml) and stably transfected P815 cells expressing Ag85. P815-HTR mastocytoma cells stably expressing Ag85A or mock-transfected were labeled with 51 Cr and used as target cells at effector:target ratios ranging from 12.5:1 to 100:1. In a separate approach, cytotoxicity was assayed by the method of Parish *et al.*³⁰.

For analysis of T-cell subsets, spleen cells were purified on Lymphocyte-M (Cedarlane) and treated with anti-CD4 (RL172 rabbit anti-mouse CD4) or anti-CD8 (83.12.5) (M. Moser, ULB) for 30 min, washed, then incubated with complement (1:9 dilution of Low-Tox, Cedarlane) for 30 min, and finally washed. This treatment was determined to substantially deplete the specific T-cell subset from the total population (~98%).

Animal challenge studies. Three to ten weeks after the last DNA injection, BALB/c and B6 mice were challenged with live mycobacteria. For BCG challenge, mice (4–6/group) were inoculated in the tail vein with 0.5 mg (2×10^6 CFU) of freshly prepared *M. bovis* BCG GL2. To enumerate BCG colony-forming units, spleens and lungs from DNA- and BCG-vaccinated mice were prepared at various times after infection, and serial threefold dilutions of the cells in PBS without antibiotics were plated on Middlebrook 7H11 Bacto Agar (Difco) and colonies were counted visually after 18 days of incubation. For statistical analysis (Student's *t*-test) data obtained from three dilutions were expressed as mean log₁₀ CFU per spleen or lungs.

To determine the resistance of the vaccinated mice to *M. tuberculosis*, mice (4 per group) were exposed in a Middlebrook airborne infection apparatus (Glas-Col, Terre Haute, Indiana) in such a manner as to deposit approximately 100 viable *M. tuberculosis* strain Erdman bacilli into the lungs. The course of the infection was then followed against time by plating serial 10-fold dilutions of organ homogenates as described above, and colonies were counted after 31 days.

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